Urdu byte-level BPE

Merge behavior, token efficiency, and diagnostic failure cases

50,000
vocabulary entries

49,743
ordered merges

Training: 375.7 s total (174.0 s counting, 201.7 s merges) | Python 3.12.15, 12 CPU cores, peak 2,329 MB; CPU model not recorded.

Model: `models/tokenizer.json`. Training used the full train split with minimum word frequency 1. Recorded host: Linux 7.2.9, x86_64.

01 What did the merges learn?

The first merges mostly reconstruct frequent two-byte Urdu letters, then combine leading spaces and letters into common word fragments. The table lists merges 1–50 and the requested 5,000-merge checkpoints; invalid byte fragments are shown in escaped form because they cannot be decoded independently.

Merge	Decoded token	Merge	Decoded token
1	ا	2	ی
3	`b'\x20\xd8'` (invalid UTF-8)	4	ک
5	`b'\x20\xd9'` (invalid UTF-8)	6	ر
7	و	8	ہ
9	ے	10	ک
11	ن	12	`b'\xd8\xa7\xd8'` (invalid UTF-8)
13	ل	14	م
15	م	16	ں
17	ا	18	ت
19	ہ	20	یں
21	س	22	د
23	`CRLF (\\r\\n)`	24	س
25	ھ	26	ب
27	گ	28	کی
29	کے	30	ب
31	پ	32	ان
33	ن	34	ور
35	ت	36	۔
37	ج	38	ار
39	`b'\x20\xd8\xa7\xd8'` (invalid UTF-8)	40	میں
41	د	42	ف
43	ق	44	ٹ
45	و	46	ع
47	ال	48	ہے
49	ائ	50	ر
5,000	پریشانی	10,000	ورحم
15,000	انوکھی	20,000	کاملہ
25,000	ٹس	30,000	منون
35,000	چکانا	40,000	میَں
45,000	atis	49,743	غپ

Twenty-one of the first 50 merges reconstruct two-byte Urdu letters. Merges 3, 5, 12, and 39 are partial UTF-8 sequences. Merge 20 first creates یں, the earliest multi-letter token that also appears as a complete corpus pre-token; “whole word” here is a corpus-pre-token criterion, not morphological analysis.

01 Twenty longest learned tokens

Ranked by UTF-8 byte length; leading spaces are part of tokens where shown.

ID	Bytes	Token	ID	Bytes	Token
26755	32	----------	35817	30	چندرابابونائیڈو
19038	29	انسائیکلوپیڈیا	36509	29	انٹرکانٹینینٹل
29505	28	نارتھیمپٹنشائر	10120	27	انانکارپوریٹڈ
18324	27	خیبرپختونخواہ	31708	27	پارلیمنٹیرینز
38710	27	انگیزانکشافات	42258	27	لاہورہائیکورٹ
49866	27	غیرجانبدارانہ	26680	26	----------
37511	26	خیبرپختونخواہ	47785	26	لاہورہائیکورٹ
7682	25	خیبرپختونخوا	23648	25	امیرالمومنین
26325	25	امیرالمؤمنین	27489	25	اورنامورمصنف
32915	25	دارالحکومتوں	37547	25	انفرااسٹرکچر

02 Tokens per whitespace-delimited word

Test-set ratios aggregate token counts and whitespace-delimited word counts. Hand examples are short diagnostics; special end-of-text markers are excluded.

Input	Words	Urdu BPE	GPT-2	o200k_base
test.txt	2,367,063	1.222	6.102	1.638
News	15	1.133	5.733	1.533
Poetry	15	1.133	6.133	1.533
Urdu with English	12	1.500	4.000	1.333
Roman Urdu	9	3.000	2.667	2.000
Numbers and dates	11	2.455	6.091	2.455
Diacritics	7	2.429	7.857	3.857
Personal names	8	1.500	7.750	2.625

On the 2.37-million-word test split, this tokenizer averages 1.222 tokens per word, compared with 6.102 for GPT-2 and 1.638 for `o200k_base`. This indicates strong compression for the source distribution, not a general quality or language-modeling score; the selected examples expose where script and spelling coverage changes the result.

03 Three inputs handled least efficiently

These are the three highest Urdu-BPE tokens-per-word ratios among the seven hand-picked inputs. Each tests a form that is less aligned with the Urdu-script word frequencies used for training.

Input type	Example	Urdu BPE / word	Why it fragments
Roman Urdu	Aaj mausam bohat khubsurat hai, chalo Islamabad chalte hain.	3.000	Romanized spellings vary and do not share the Urdu-script merges learned by this tokenizer.
Numbers and dates	قیمت ۱۲۵۰ روپے ہے؛ تاریخ 2026-10-11 اور وقت 9:45 بجے ہے۔	2.455	Mixed numeral conventions, punctuation, and date formats produce combinations that are not reusable word-like units.
Diacritics	اَللّٰہُ أَکْبَرُ، اُردُو کے اعراب اور حرکات	2.429	Combining marks add UTF-8 bytes and are relatively sparse, so marked words remain more fragmented.

Interpretation and limits

The low test-split ratio reflects the tokenizer's fit to a large Urdu-script corpus, while Roman Urdu is the clearest weakness in the selected set. Digit normalization is applied when preparing the corpus, but the hand-written diagnostic retains mixed digit forms, so its result also tests normalization mismatch.

The three failure cases are maxima from seven manually selected inputs, not a representative benchmark. Tokens-per-word is a compression measure only; it does not evaluate semantic quality, downstream model performance, or fairness across Urdu varieties.

Urdu-aware pre-tokenization

```
's|'t|'re|'ve|'m|'ll|'d| ?[\p{L}\p{M}\u200c]+| ?\p{N}+| ?[^\s\p{L}\p{M}\p{N}\u200c]+|\s+(?!\S)|\s+
```

Alternative	Behavior and reason
`'s|'t|'re|'ve|'m|'ll|'d`	Keep common English contractions as separate pieces for naturally code-mixed Urdu.
`?[\p{L}\p{M}\u200c]+`	Group Unicode letters, combining marks, and ZWNJ. Marks and joiners stay inside the pre-token, so they do not create artificial BPE boundaries. `\p{L}` and `\p{M}` cover Urdu and other scripts.
`?\p{N}+`	Group a digit run independently from words; the optional leading ASCII space remains attached to it.
`?[^\s\p{L}\p{M}\p{N}\u200c]+`	Group runs of punctuation or symbols, excluding whitespace, letters, marks, digits, and ZWNJ; this avoids splitting punctuation into single-character pieces.
`\s+(?!\S)|\s+`	Consume whitespace runs, preferring trailing whitespace as one piece and then matching any remaining run. Together, the alternatives cover newlines without losing them.

Because BPE counts pairs within each pre-token, no learned merge crosses these boundaries. The normalization step is separate; this pattern preserves ZWNJ rather than removing it.

Analysis generated from `models/tokenizer.json`, `data/processed/test.txt`, and seven fixed diagnostic strings. Comparison encodings: `tiktoken` GPT-2 and `o200k_base`. Merge indices are one-based; token IDs follow the 256-byte base vocabulary and special ID 256.